

Department of Computer Engineering COMSATS

University Islamabad, Attock Campus

Lab Rubrics Form	
Lab Name	Signal Operations: Time Reversal and Expansion
Lab Number	05
Submitted by:	
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Rubrics		Marks		
		1	2	3
PLO-5 Modern Tools Usage	Ability to execute the experiment utilizing hardware, software tools, or a combination of both.			
PLO-10 Communication Skills	Present lab activities effectively through detailed lab reports, oral examinations, and engaging presentations.			

Lab-05

Signal Operations: Time Reversal and Expansion

Objective:

The objective of this lab is to understand and implement fundamental discrete-time signal operations, specifically:

1. **Time Reversal:** To reverse the discrete-time signal $x[n]$ in time, producing $x[-n]$.
2. **Time Expansion:** To expand the signal in time by a factor M , producing $x[n/M]$, and to observe the effect of linear interpolation when required.

Lab Task:

For a discrete time signal defined below:

$$x[n] = \begin{cases} e^{-n}, & \text{if } -3 \leq n < -1 \\ 1, & \text{if } 0 \leq n \leq 6 \\ 0, & \text{otherwise} \end{cases}$$

1. Plot the signal.
 2. Time Reversal.
 3. Time Expansion.
- Perform time expansion operation.
 - Plot and discuss your results.
 - Use linear interpolation wherever necessary.

Code:

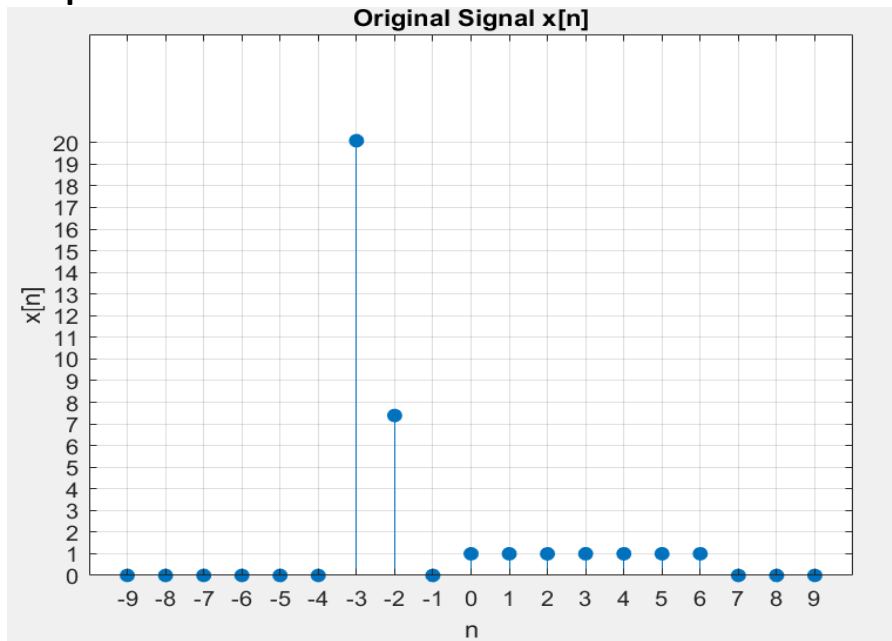
```
clear; close all;
n = -9:9;
x = zeros(size(n));
for k = 1:length(n)
    if n(k) >= -3 && n(k) < -1
        x(k) = exp(-n(k));
    elseif n(k) >= 0 && n(k) <= 6
        x(k) = 1;
    else
        x(k) = 0;
    end
end
figure;
stem(n, x, 'filled');
```

```

title('Original Signal x[n]');
xlabel('n');
ylabel('x[n]');
grid on;
xticks(n);
yticks(min(x):1:max(x));

```

Output:



Code:

```

clc; clear; close all;
% Define n range
n = -9:9;
x = zeros(size(n));
% Define the signal x[n]
for k = 1:length(n)
    if n(k) >= -3 && n(k) < -1
        x(k) = exp(-n(k));
    elseif n(k) >= 0 && n(k) <= 6
        x(k) = 1;
    else
        x(k) = 0;
    end
end
% Time reversal: y[n] = x[-n]
y = zeros(size(n));
for i = 1:length(n)
    idx = find(n == -n(i), 1);

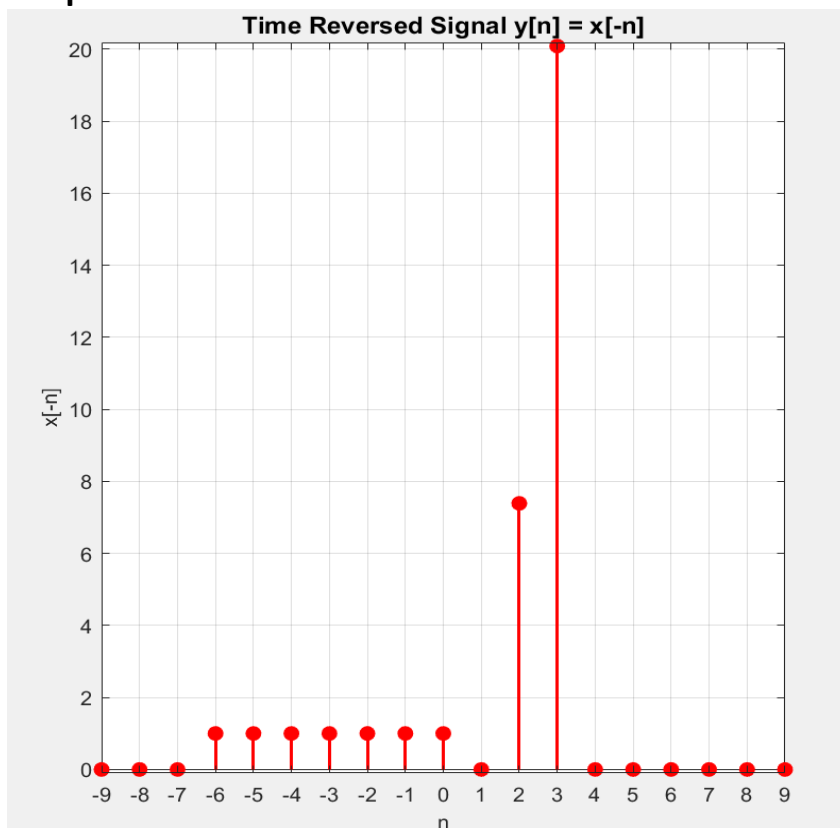
```

```

    if ~isempty(idx)
        y(i) = x(idx);
    end
end
% Plot time reversed signal
figure;
stem(n, y, 'r', 'filled', 'LineWidth', 1.5, 'MarkerSize', 6);
title('Time Reversed Signal x[-n]');
xlabel('n');
ylabel('x[-n]');
grid on;
xlim([-9, 9]);
ylim([-0.1, max(y)+0.1]);
xticks(n);
yticks(y);

```

Output:



Code:

```

% Time Expansion (M = 2)
M = 2;           % Expansion factor
n_exp = -9:9;    % New index range
x_exp = zeros(size(n_exp)); % Initialize expanded signal

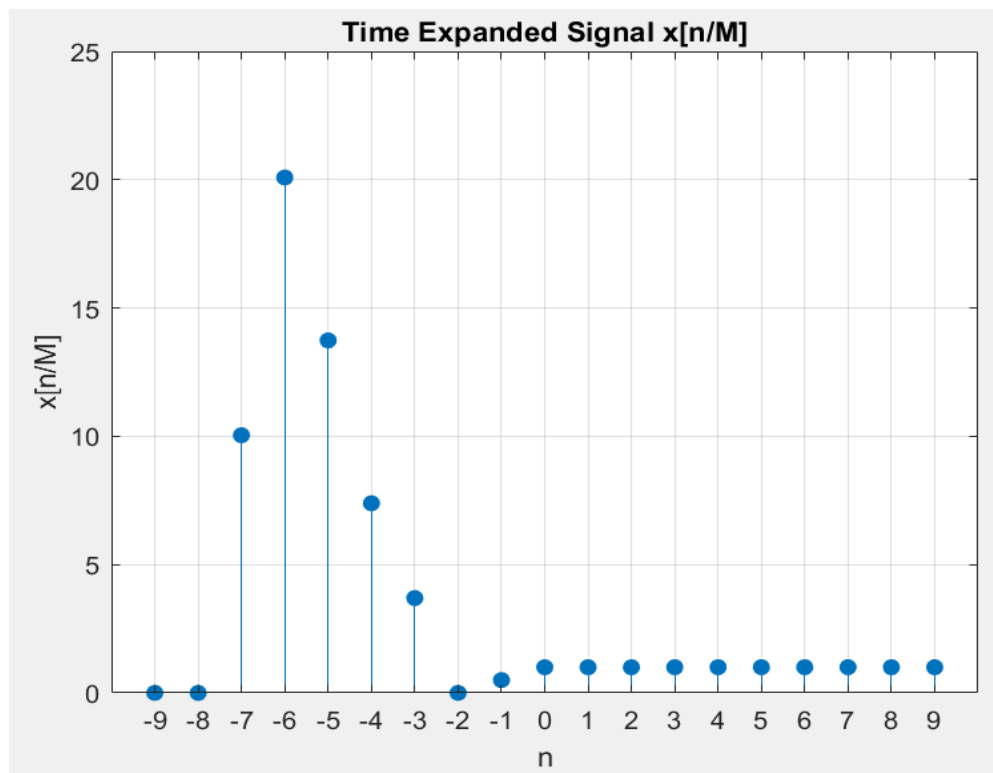
```

```

for i = 1:length(n_exp)
    t = n_exp(i)/M;      % Map expanded index to original
    if mod(t,1) == 0     % If t is integer, take exact sample
        idx = find(n == t);
        if ~isempty(idx)
            x_exp(i) = x(idx);
        end
    else                 % Linear interpolation for non-integer t
        n1 = floor(t);
        n2 = ceil(t);
        if any(n==n1) && any(n==n2)
            x1 = x(n==n1);
            x2 = x(n==n2);
            x_exp(i) = x1 + (t-n1)*(x2-x1);
        end
    end
end
% Plot
figure;
stem(n_exp, x_exp, 'filled');
title('Time Expanded Signal x[n/M]');
xlabel('n');
ylabel('x[n/M]');
grid on;
xticks(min(n_exp):1:max(n_exp));

```

Output:



Conclusion:

In this lab, we successfully implemented and visualized time reversal and time expansion operations on a discrete-time signal using MATLAB.

- The **time reversal** operation correctly reflected the signal about the vertical axis, as shown in the plot of $x[-n]$.
- The **time expansion** operation ($M=2$) stretched the signal in time, with missing samples filled using linear interpolation, resulting in a smoother expanded signal.